

WITTERING, United Kingdom

WMO: 034620

Lat: 52.6114N

Lon: 0.4611W

Elev: 83

StdP: 100.33

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.5	(c) -2.1	(d) -6.1	(e) 2.3	(f) -1.0	(g) -4.8	(h) 2.6	(i) -0.1	(j) 15.2	(k) 9.2	(l) 13.9	(m) 9.2	(n) 3.2	(o) 270	(p) 0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.2	(c) 26.7	(d) 18.1	(e) 24.5	(f) 17.3	(g) 22.7	(h) 16.6	(i) 19.2	(j) 24.3	(k) 18.2	(l) 22.7	(m) 17.4	(n) 21.4	(o) 4.5	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	12.6	20.8	16.5	11.9	19.8	15.7	11.2	19.0	54.9	24.5	51.7	22.8	49.1	21.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.4	(b) 11.0	(c) 9.8	(e) -6.3	(f) 30.6	(g) 2.3	(h) 1.8	(i) -8.0	(j) 31.9	(k) -9.3	(l) 33.0	(m) -10.6	(n) 34.0	(o) -12.3	(p) 35.4		
(a) 12.4	(b) 11.0	(c) 9.8	(e) -6.6	(f) 21.4	(g) 2.3	(h) 1.8	(i) -8.3	(j) 22.7	(k) -9.6	(l) 23.8	(m) -10.9	(n) 24.8	(o) -12.6	(p) 26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	4.3	4.6	6.4	8.7	11.6	14.6	17.0	16.8	14.3	10.9	7.1	4.7
	DBStd	5.44	3.15	3.25	2.98	2.95	2.90	2.64	2.60	2.50	2.36	2.83	3.03	3.60
	HDD10.0	806	177	152	117	58	16	1	0	0	1	22	96	167
	HDD18.3	3036	435	383	371	288	209	116	58	61	124	229	338	423
	CDD10.0	851	1	2	4	20	66	139	218	212	129	51	7	3
	CDD18.3	39	0	0	0	0	0	4	17	15	2	1	0	0
	CDH23.3	321	0	0	0	1	4	43	144	110	15	2	0	0
	CDH26.7	62	0	0	0	0	0	7	29	24	2	0	0	0
Wind	WSAvg	4.9	5.6	5.5	5.3	4.8	4.7	4.4	4.3	4.3	4.5	4.8	4.9	5.3
Precipitation	PrecAvg	580	45	35	47	49	54	51	47	57	44	46	53	51
	PrecMax	748	87	117	97	85	133	154	128	114	107	129	130	130
	PrecMin	426	14	12	9	11	16	3	6	3	8	4	21	7
	PrecStd	74	20	23	26	21	32	36	30	29	27	34	24	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.0	16.4	21.2	23.8	27.8	29.9	29.4	25.5	20.4	15.5	13.8
		MCWB	11.1	11.4	11.0	14.5	16.9	19.1	19.2	18.6	17.2	16.0	13.6	11.9
	2%	DB	11.6	12.1	14.2	17.7	21.4	24.2	26.8	26.4	22.4	17.9	14.1	12.6
		MCWB	10.0	9.8	10.3	12.1	15.4	16.8	17.9	18.4	16.5	14.6	12.5	11.2
	5%	DB	10.3	10.7	12.7	15.8	19.2	22.0	24.7	23.7	20.3	16.4	12.9	11.4
		MCWB	8.8	8.7	9.5	11.0	14.0	16.0	17.4	17.2	15.4	13.9	11.3	10.1
	10%	DB	9.1	9.6	11.4	14.0	17.1	20.0	22.7	21.9	18.8	15.2	11.7	10.1
		MCWB	7.8	7.9	8.7	10.1	12.7	14.9	16.6	16.3	14.5	13.1	10.3	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	11.6	12.3	14.9	18.0	20.3	20.6	20.3	18.7	16.6	14.0	12.2
		MCDB	12.7	13.4	14.7	20.2	22.4	26.1	27.3	26.4	22.8	19.4	15.0	13.1
	2%	WB	10.2	10.1	11.1	13.0	15.9	18.0	19.2	19.0	17.3	15.5	12.7	11.4
		MCDB	11.4	11.7	13.4	16.9	20.3	22.0	24.5	23.9	20.7	17.4	13.8	12.4
	5%	WB	9.1	9.0	10.0	11.7	14.6	16.7	18.2	18.0	16.3	14.5	11.6	10.2
		MCDB	10.1	10.5	12.2	14.7	18.1	20.6	22.9	22.4	19.0	16.0	12.6	11.3
	10%	WB	8.0	8.0	9.0	10.5	13.4	15.7	17.2	17.1	15.4	13.5	10.5	9.0
		MCDB	9.0	9.4	11.0	13.4	16.5	19.3	21.6	20.9	17.8	15.0	11.5	10.0
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.9	7.0	8.3	8.6	8.7	9.2	8.9	8.1	6.7	5.6	4.9
		MCDBR	6.4	6.7	9.0	10.9	11.9	12.0	12.8	11.8	10.4	7.5	6.1	5.8
		MCWBR	5.8	5.3	5.9	6.3	6.8	6.1	5.4	5.1	5.4	5.0	5.2	5.4
	5% WB	MCDBR	6.4	6.5	8.1	9.9	10.8	10.6	11.1	10.4	9.0	7.1	6.0	5.7
		MCWBR	6.0	5.4	5.8	6.2	6.6	6.2	5.5	5.0	5.5	5.3	5.3	5.4
Clear-Sky Solar Irradiance	taub		0.322	0.338	0.370	0.396	0.400	0.403	0.402	0.390	0.370	0.351	0.325	0.311
	taud		2.365	2.388	2.291	2.229	2.259	2.273	2.283	2.341	2.402	2.438	2.415	2.344
	Ebn at Noon		664	771	811	835	851	850	843	833	807	745	657	602
	Edh at Noon		60	78	104	125	127	127	123	110	93	74	57	52
All-Sky Solar Radiation	RadAvg		0.74	1.44	2.51	3.86	4.77	5.08	4.89	4.04	3.02	1.75	0.92	0.61
	RadStd		0.07	0.17	0.30	0.42	0.31	0.49	0.53	0.28	0.27	0.15	0.10	0.06

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (91 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page